National Institute of Open Schooling
Senior Secondary Course
Lesson 35 : Plane
Worksheet- 35

1. Take (a, b, c) be the coordinate of point P on the plane and l, m, and n be the direction cosines. Find the equation of a plane in vector form and normal form.
2. Find the vector equation of a plane passing through the point (4, -6, 5) and perpendicular to the line with direction ratios 3, 2, -1.
3. If the plane passing through three points X (3, 5, -2), Y (4, -3, 5), and (-4, 3, 5), then find the vector equation of the plane.
4. The equation of a plane is 2x + 4y -5z – 40 = 0. Reduced the equation of a plane to the intercept form and find its intercepts on the co-ordinate axes.
5. If the points A (1, 1, a) and B (–3, 0, 1) are equidistant from the plane 3x+4y–12z+13=0, Find the value of a.
6. Find the equation of the plane passing through the points (2,–1,5), and perpendicular to the planes x+2y–z =1 and 3x –4y +z = 5.
7. Find the equation of the plane parallel to the plane 2x + 3y – 6z –5 = 0 and passing through the point (2, 1, –3).
8. Find the equation of the plane passing through the origin and perpendicular to the planes 2x + 2y + 2z = 0 and 2x + 3y – 2z=0
9. Find the equation of the plane passing through the points A (1, - 2, 3), B(3,–1,2) and parallel to the lines $\frac{x-4}{1}=\frac{y+3}{-4}=\frac{z+1}{7}$
10. Find the distance between the planes 2x + 3y – 2z – 15 = 0 and 4x + 6y – 4z - 30=0